UUM
Universiti Utara Malaysia

FINAL EXAMINATION
FIRST SEMESTER 2023/2024 SESSION

COURSE CODE/NAME	:	STIW3034 / SOFTWARE TESTING AND QUALITY ASSURANCE
DATE	:	24 FEBRUARY 2024 (SATURDAY)
TIME	:	2.30 P.M.- 5.00 P.M. (2 ½ HOURS)
VENUE	:	DTSO

INSTRUCTIONS:

1. This examination paper has **TWO (2) SECTIONS, A** and **B**, in **EIGHTEEN (18)** printed pages excluding the cover page. **SECTION A** has **TWENTY-FOUR (24)** multiple choice questions and **SECTION B** has **ELEVEN (11)** structured questions.
2. Answer **ALL** questions in the space provided.
3. You are **NOT ALLOWED** to remove the examination paper from the examination hall.

MATRIC NO.: _____ (in words) | | | | | | | (in numbers)

IDENTIFICATION CARD NO.: | | | | | | | | | | | | |

LECTURER NAME: _____

GROUP: | | **TABLE NO.:** | | | | |

DO NOT OPEN THIS EXAMINATION PAPER UNTIL INSTRUCTED

SECTION A: MULTIPLE CHOICE QUESTIONS (24 MARKS)

INSTRUCTION: Answer **ALL** questions in the space provided.

1. A company recently bought a commercial shelf-shopping application to automate their company's buying and selling processes. They plan to run an acceptance test against the package before putting it into production. Which of the following is their most likely reason for testing?

   A. To build confidence in the application
   B. To detect bugs in the applications.
   C. To gather evidence for a lawsuit.
   D. To train the users.

2. One of the differences between the Modified Condition Decision Coverage (MCDC) and the Multiple Condition Coverage (MCC) is _____.

   A. The MCC ensures all paths through a module are executed, whereas the MCDC ensures each path is a decision that can independently affect the outcome.
   B. The MCDC ensures all alternative paths are executed, whereas the MCC ensures all main paths are executed.
   C. The MCC ensures both conditions take all possible outcomes at least once, whereas the MCDC requires maximum and minimum values within each field.
   D. The MCDC relies on the testers' skill and experience, whereas the MCC relies on the way in which the system moves from one condition to another.

3. Which of the following annotations causes that method to be run after each test method?

   A. @Test
   B. @ Before
   C. @After
   D. @AfterClass

4. Which method is used to verify the actual and expected results in JUnits?

   A. equals()
   B. assertThat()
   C. isEqual()
   D. ==

5. Which of the following is correct about Test Suite in JUnit?

   A. @RunWith annotation is used to run the suite test.
   B. Test suite means bundling a few unit test cases and running them together.
   C. @Suite annotation is used to run the suite test.
   D. All of the mentioned.

6. Which annotation implies that a method is a JUnit test case?

   A. @org.junit.Test
   B. @org.Test
   C. @testcase
   D. @junit

7. Which of the following is among the typical tasks of a test leader?

   A. Develop system requirements, design specifications, and usage models.
   B. Handle all test automation duties.
   C. Keep tests and test coverage is hidden from programmers.
   D. Gather and report test coverage.

8. What do we call a document that describes any event during testing requiring further investigation?

   A. A bug report.
   B. A defect report.
   C. An incident report.
   D. A test summary report.

9. Which of the following should contain the test summary report?

   A. Performance appraisal of the development team.
   B. List of process improvement activities to be taken.
   C. Detailed schedule of the maintenance releases.
   D. Deviations from planned testing.

10. Which one of the following would NOT typically be covered in a test summary?

    A. An analysis of lessons learned to determine changes for future projects.
    B. Any features of the system that were not tested along with reasons.
    C. The risk associated with unresolved defects found during testing.
    D. Whether there is any economic benefit in continuing testing beyond the planned date.

11. A product risk analysis is performed during the planning stage of the test process. During the execution stage of the test process, the test manager directs the testers to classify each defect report by the known product risk it relates to or to "others". Once a week, the test manager runs a report showing the percentage of defects related to each known product risk and unknown risks. What is one possible use of such a report?

    A. To identify new risks to system quality.
    B. To locate defect clusters in the product subsystem.
    C. To check risk coverage by tests.
    D. To measure exploratory testing.

12. Which of the following is **NOT** a part of Configuration Management?

    A. Controlled library access.
    B. Record of changes to documentation over time.
    C. Auditing conformance to ISO9001.
    D. Identification of test versions.

13. Which of the following **BEST** describes a keyword-driven testing approach?

    A. Test input and expected results are stored in a table or a spreadsheet so that a single control script can execute all the tests in the table.
    B. Action words are defined to cover specific interactions in the system (for example, log-on entries), which testers can then use to build their tests.
    C. Keywords are entered into a test harness to obtain pre-defined test coverage reports.
    D. The test basis is searched using keywords to help suitable test conditions.

14. Which of the following metrics would be most useful to monitor during test execution?

    A. Number of test environments remaining to be configured.
    B. Percentage of the requirement for which a test has been written.
    C. Percentage of test cases written.
    D. Number of defects found and fixed.

15. Which one of the following will most probably influence the duration but not the effort of the testing activities?

    A. Required quality of the system.
    B. Required level of detail of test conditions.
    C. Maturity of the test processes.
    D. Time to repair defects found.

16. Which one of the following would be a reason to collect efficiency metrics in reviews?

    A. To determine the competency of the reviewers.
    B. To appraise the performance of the review leader.
    C. To assess the efficiency of the author of the work product.
    D. To determine if the review was a good use of resources.

17. Your management is complaining that too much time is being spent on requirements reviews and would like to stop the practice. You think the reviews are very valuable and have significantly improved the quality of the code due to the removal of requirements defects found during the reviews. Which of the following metrics would be most useful in convincing your management that the reviews should continue?

    A. Time required to prepare for and conduct the review.
    B. Number of defects found and their severity.
    C. Defect detection effectiveness and the time required to fix the defects found.
    D. Cost of quality metrics for the defects found in the reviews versus other methods.

18. Which one of the following statements about testing metrics usage is **TRUE?**

    A. Trends in the lag time from defect reporting to resolution are used to reward the developers.
    B. The number of test conditions identified is used to monitor testing quality.
    C. Planned versus actual hours to develop testware are used to minimise regression testing.
    D. Confirmation and regression test status are used to monitor testing progress.

19. Which one of the following alternatives is best to monitor test progress?

    A. Combined usage of coverage, confidence, risk, test, and defect metrics.
    B. Estimation of code coverage by measuring the number of executed tests.
    C. Combined usage of development, product, people, test, and defect metrics.
    D. Combined usage of the product, people, and project metrics.

20. Which of the following is a key characteristic of a Walkthrough?

    A. The author leads the meeting.
    B. Usually led by a trained manager.
    C. Formal follow-up process.
    D. Includes strict exit criteria.

21. Which of the following is **FALSE** about a Formal Review or Inspection?

    A. The meeting is led by a trained moderator.
    B. Individual preparation before the review meeting is optional.
    C. There is a formal follow-up process.
    D. The author participates in the review meeting.

22. The phases of a formal review process are mentioned below.

    i. Planning
    ii. Review meeting
    iii. Rework
    iv. Individual preparation
    v. Initiating
    vi. Follow up

    Arrange them in the correct order.

    A. i, iii, ii, iv, vi, v
    B. vi,i, ii, iii, iv, v
    C. i, v, iv ii, iii, vi
    D. v, i, ii, iv, iii, vi

23. Which of the following is **NOT** a role in a Formal Review process?

A. Manager
B. Creator
C. Moderator
D. Scriber

24. Static code analysis typically identifies all, **EXCEPT**_____.

A. Unreachable code
B. Undeclared variables
C. Faults in the requirements
D. Too few comments

SECTION B: STRUCTURED QUESTIONS (76 MARKS)

INSTRUCTION: Answer **ALL** questions in the space provided.

1. Identify **FOUR (4)** testing principles. For each principle, provide an example that relates to it.

(8 marks)

2. Explain how test level and test type are related. Provide some examples to support your elaboration.

(4 marks)

3. State **TWO (2)** reasons why error guessing is a useful design technique.

(4 marks)

4. You have received the following test execution completion information from your team. The exit criteria state that you must have an 80% pass rate of all test cases and an 85% pass rate of all test cases covering high-risk areas (risk priority of 1 or 2).

Risk Area	Risk Priority	Test Passed	Test Failed	Test not Executed
1	1	8	2	0
2	3	10	0	2
3	4	5	0	0
4	2	7	3	2
5	3	10	0	0
6	3	10	0	0
7	5	8	0	10
8	1	5	1	2
9	2	20	0	0
10	2	7	2	1

a) Does this system testing satisfy its exit criteria?

(1 mark)

b) Based on your decision a) provide your justification.

(3 marks)

5. Analyse the following conditions. An automated air-conditioner is programmed to turn its heating unit when the temperature falls below 17°C and turn its refrigeration unit on when it exceeds 26 °C. The air-conditioner is designed to operate at temperatures between -10 °C and 40 °C. Given the above specification, break down these conditions into several classes. Suggest a set of values using the equivalence partition test design technique. Include valid and invalid classes.

(10 marks)

6. ShopMore website shopping cart starts empty. When a product is selected, it is added to the shopping cart. Items can be removed from the shopping cart. When the customer decides to check out, a summary of the items in the cart and the total cost are shown, for the customer to say whether this is okay or not. If the content and price are OK, then you leave the summary display and go to the payment system. Otherwise, you go back to shop (so you can remove items if you want).

a) Produce a state- diagram showing the different states and transitions.

(8 marks)

b) Identify the 0-switch for this coverage for this state transition diagram.

(4 marks)

7. Analyse the following scenario, and recommend **TWO (2)** types of testing activities that you can perform for the payment web-based application. For each activity, include some explanation.

For our financial web-based application, assessments encompass rigorous checks on data encryption, resistance against vulnerabilities like cross-site scripting, and authentication robustness. Meanwhile, system responsiveness, scalability during peak loads, and swift server responses across various platforms need to be conducted. These measures fortify data integrity, safeguard against cyber threats, and ensure uninterrupted financial transactions. A meticulous evaluation of these aspects guarantees a secure, resilient, and high-performance application, instilling trust in users and facilitating seamless financial operations.

(4 marks)

8. You are organising the test execution for software that will be used to monitor the amount of insulin that is released by a pump. This software is quite complex and tracks the patient's vital signs and blood sugar to determine the amount of insulin needed. This is safety-critical software and is subjected to regulatory requirements. No release, not even a beta version, can be conducted unless all regulatory requirements have been met. There are features that the business thinks are critical to the success of the project. These include some new reporting capabilities that will help to differentiate this product from the competition.
You have created the following table of the test cases to be executed. 1 is the highest priority. Sprints are two weeks long and are numbered sequentially.

Test case ID	Risk Priority	Business Priority	Regulatory	Software Availability
1	Very High	High	No	Sprint 4
2	High	Medium	Yes	Sprint 3
3	Medium	High	No	Sprint 2
4	High	High	Yes	Sprint 1
5	Low	Medium	Yes	Sprint 2
6	Medium	Low	Yes	Sprint 3
7	Very High	Low	Yes	Sprint 3

Assess the above scenario, list the proper order for test execution in the table below.

(8 marks)

9. In developing critical healthcare management software, a risk arises due to a potential software flaw leading to harmful consequences if misinterpreted medical data alters patient treatment plans. Simultaneously, the software faces the risk of not meeting its intended functions, causing inefficiencies in diagnosis or treatment protocols. A shortage of skilled developers amplifies these risks, delaying crucial error detection and resolution. Additionally, ambiguities in defining software requirements create a challenge, potentially leading to misaligned development efforts.

Assess the above scenario, and list **TWO (2)** risk types involved. For each risk type identified, problem elaboration.

(6 marks)

10. Identify **TWO (2)** ways configuration management can support better software testing.

(4 marks)

11. Read the following scenario.

A software development team at FirmTech encountered persistent anomalies in a financial application. Despite thorough coding and testing, discrepancies persisted in transaction calculations. Recognising the issue, the team collectively reviewed the code and identified irregularities in the algorithm's logic during a routine code review session. Discussions ensued, outlining the defect's nature as an arithmetic calculation error. The team effectively classified the defect's type and rectified it promptly. This incident underscored the significance of comprehensive software review practices in identifying and addressing intricate coding discrepancies, enhancing overall software quality.

Evaluate the above scenario, suggest **TWO (2)** relevant standards that could be applied by this company to systematically tackle their current problem. For each chosen standard provide, support with a justification and example.

(12 marks)

ANSWERING SPACE FOR QUESTION 11.

END OF QUESTIONS